

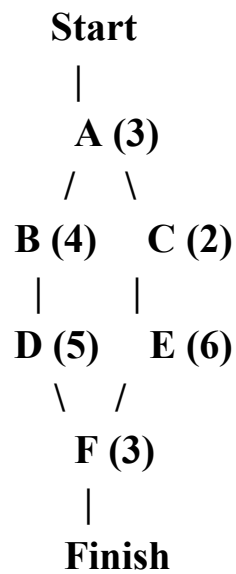
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Course Name: Project Management

Assignment: Critical Path Method

Project Network Diagram:



Earliest Start and Earliest Finish for each activity

Activity A

- $ES = 0$
- $EF = 0 + 3 = 3$

Activity B (after A)

- $ES = 3$
- $EF = 3 + 4 = 7$

Activity C (after A)

- $ES = 3$
- $EF = 3 + 2 = 5$

Activity D (after B)

- $ES = 7$
- $EF = 7 + 5 = 12$

Activity E (after C)

- $ES = 5$
- $EF = 5 + 6 = 11$

Activity F (after D and E)

- $ES = \max(12, 11) = 12$
- $EF = 12 + 3 = 15$

Identify the critical path:

Critical path = longest path

Check paths:

- $A \rightarrow B \rightarrow D \rightarrow F = 3 + 4 + 5 + 3 = 15$
- $A \rightarrow C \rightarrow E \rightarrow F = 3 + 2 + 6 + 3 = 14$

So:

Critical Path = $A \rightarrow B \rightarrow D \rightarrow F$

Total Project Duration

15 days